Introduction to Artificial Intelligence

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Assignment:
NLP – text classification

In this lab session, you implement NLP text classification for predicting a genre from movie overview texts. You will be working in your teams of two students each.

1. Read Sections 6.1 and 6.2 of the book “Applied Artificial Intelligence – An Engineering Approach”.
2. Answer the quick check questions
3. Download Movie_Overview_Classification.csv from the Moodle course
4. In your Pycharm project, create a new Python package “05a_NLP_text_classification”
5. Create a new subfolder data and store Download Movie_Overview_Classification.csv
6. Create a new Jupyter notebook NLP_classifyer.ipynb
7. Implement the classifier
    a. Load data
    b. Perform pre-processing, e.g., replacing missing values if existing
    c. Implement a classifier
    d. Measure accuracy using cross-validation
    e. Optimize your implementation by modifying the classifier and / or pre-processing
8. Commit your code to the Git repository
9. Enter your accuracy score in the leaderboard in the shared spreadsheet.